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Chapter 3

Professional Relationships

Relationship	Foundation	Professional's Main Duties	Example
Employer – Employee	Contractual relationship	Loyalty, honesty, confidentiality, don't reveal trade secrets	Employee signs NDA
Client – Professional	Contractual relationship	Deliver agreed work ethically	Software development contract
Society – Professional	Public trust & law	Due care, don't harm society, protect public	Safe medical software
Professional – Professional	Professional standards	Honesty, loyalty, don't take bribes or falsify results	Engineer reports qualifications honestly

Models of Client–Professional Relationships

Model	Who Makes Decisions?	Professional's Role	Client's Role	Example
Paternalistic Model	Professional	Makes all decisions	Accepts decisions	Doctor decides treatment
Trust Model	Shared	Provides options and advice	Chooses among options	Software consultant recommends solutions
Agency Model	Client	Carries out client's instructions	Makes decisions	Stockbroker buys requested shares

Employer–Employee Responsibilities

Employer Responsibilities	Employee Responsibilities
Fair wages	Loyalty
Safe workplace	Honesty
Respect employees	Protect trade secrets
Don't exploit workers	Follow employment contract
Honor agreements	Don't work for competitors if restricted

Society–Professional Responsibilities

Responsibility	Purpose	Example
Serve Society	Benefit the public	Build reliable healthcare software
Do No Harm	Prevent damage	Don't release unsafe software
Due Care	Apply professional expertise carefully	Thorough testing
Consider End Users	Protect everyone affected	Design accessible systems
Protect Society Through Legislation	Follow laws and standards	Data protection compliance

Whistle-Blowing Process

Stage	Description	Goal
Internal Reporting	Report inside the organization	Solve the issue quietly
External Whistle-Blowing	Report to media or government after internal reporting fails	Protect the public
Whistle-Blower	Person reporting illegal or unethical behavior	Stop harmful practices

Internal Reporting vs External Whistle-Blowing

Feature	Internal Reporting	External Whistle-Blowing
Reported To	Managers or organization	Media or government
Publicity	Usually none	High
Preferred First?	Yes	Only if internal reporting fails
Purpose	Fix the problem internally	Expose unresolved wrongdoing
Organizational Impact	Minimal	Potentially severe

Whistle-Blowing Overview

Topic	Details
Factors That Inhibit Internal Reporting	Unavailable resources, hostile culture, toxic leadership, lack of organizational justice
Good Motives	Help the public
Questionable Motives	Revenge, retaliation, avoiding punishment
Types of Whistle-Blowers	Altruist, Avenger, Organization Man, Alarmist, Bounty Hunter

Corporate Response to Whistle-Blowing

Impact	Organization	Whistle-Blower
Reputation	Bad publicity	Labeled as troublemaker
Workplace	Loss of trust and teamwork	Isolation from coworkers
Career	Managers' careers may suffer	Retaliation
Future	Legal consequences	Poor career prospects

Types of Responsibility

Responsibility	Definition	Exclusive?	Example
Role Responsibility	Responsibility based on assigned job duties	Yes	Bookkeeper pays bills
Causal Responsibility	Responsibility because your actions caused an event	Yes	Employee releases virus
Legal Responsibility	Responsibility imposed by law	Yes	Homeowner liable for injuries
Moral Responsibility	Ethical responsibility toward others	No	Entire software team responsible for safety

Professional Codes of Ethics

Topic	Key Points
Purpose	Defines principles and core values of a profession
Why We Need It	Ethical starting point, ensures quality, guides ethical decisions
What It Is NOT	Not law, not a complete ethical framework, not a checklist
Functions	Inspiration, Education, Guidance, Accountability, Enforcement, Decision-Making Tool

Chapter 4

Problems Faced by Systems Analysts & Software Engineers

Problem	Description	Consequence	Example
Software Specification	Incomplete or unclear requirements	Incorrect or missing functionality	Customer requirements are incomplete
Software Design	Communication gap between domain experts and programmers	Design mistakes	Brake engineer explains requirements incorrectly
Extreme Pressure	Pressure to release products quickly	Bugs and poor quality	Shipping software before testing
Unauthorized Software	Using illegal software or uncredited open-source code	Legal and maintenance problems	Pirated IDE or copied GitHub code
Software Maintenance	Modifying software after deployment	Small changes may reduce reliability	Bug fix introduces another bug
Insufficient/Incompetent Staff	Poor planning, weak management, lack of experienced staff	Low-quality software	No senior developer on project

Common Ethical Problems During Software Development

Ethical Problem	Description	Main Risk	Example
Red Lies	Knowingly giving false project status	Loss of client trust	Saying the project is on schedule when it isn't
Sweep it Under the Rug	Ignoring problems instead of fixing them	Bigger failures later	Ignoring a security bug
Cancelled Vacation Syndrome	Forcing employees to sacrifice personal time	Employee burnout	Canceling vacations before release
Extreme Pressure	Prioritizing deadlines over quality	Unsafe software	Skipping testing
Compromising Quality	Releasing software despite known defects	Harm to users	Launching an unstable system

Methods for Preventing Software Development Problems

Method	Purpose	Benefit	Example
Software Development Methodology	Standardize development	Reduces errors	Agile / Waterfall
Quality Assurance (QA)	Verify product quality	Reliable software	QA testing before release
Documentation	Record system details	Easier maintenance	Requirements document
Project Management	Improve planning and estimation	Better resource allocation	Timeline planning
Professional Ethics	Guide decisions	Prevent unethical behavior	Refusing to release unsafe software

Development of Safety-Critical Systems

Practice	Purpose	Why It's Important	Example
Rigorous Development Process	Careful software development	Reduce failures	Medical software
Documentation	Record every requirement and change	Traceability	Aircraft software
Checking & Rechecking	Verify every component	Detect hidden errors	Nuclear reactor software
System Safety Engineer	Monitor hazards throughout project	Improve safety	Aviation industry
Hazard Log	Record and track hazards	Ensure hazards are resolved	Risk register
Testing	Verify sufficient testing	Prevent catastrophic failures	Brake system testing
Risk Analysis	Analyze possible failures	Reduce risk	Identify possible aircraft failures
Redundancy	Multiple interchangeable components	Continue operation after failure	Dual aircraft computers
N-Version Programming	Multiple independent implementations	Reduce software errors	Three flight control programs with voting algorithm
Emergency Procedures	Prepare for failures	Reduce damage	Emergency shutdown

Risk Analysis Components

Component	Question Answered	Example
Hazard Identification	What can go wrong?	Engine failure
Likelihood	How likely is it?	1 in 1 million
Consequences	What happens if it occurs?	Loss of life
Prevention	How can it be prevented?	Better design
Detection	How can it be detected early?	Warning system
Mitigation	How can damage be reduced?	Backup system

Professional Responsibilities of Systems Analysts & Software Engineers

Responsibility	Description	Example
Confidentiality	Protect employee information	HR database
Corporate Confidentiality	Protect company secrets	Source code
Honesty	Admit mistakes	Report software bug
Communication	Be open with stakeholders	Inform client of delays
Follow-through	Complete assigned work	Finish documentation
Self-Assessment	Know personal strengths and weaknesses	Ask for help when needed

Sources of Ethical Guidance

Source	Purpose	Example
Professional Bodies	Publish codes of ethics	IEEE, ACM
Information Systems Methodologies	Provide structured development process	SDLC
Deontological Approach	Follow duties and rules	Never lie to clients

Responsibilities of an IT System Administrator

Responsibility	Purpose
User Administration	Manage user accounts
Monitor System Performance	Ensure systems run efficiently
Install & Update Software	Keep systems current
Monitor Network Communication	Detect suspicious activity
Implement Security Policies	Protect organizational assets
Backup & Recovery	Restore lost information
Password & Identity Management	Secure authentication
Cloud Infrastructure Management	Maintain cloud services

Ethical Issues Faced by System Administrators

Ethical Issue	Description	Example
Invasion of Privacy	Monitoring employee activity	Reading employee emails
Equality in Reporting	Reporting violations fairly	Reporting every employee equally
Sensitive Information	Protect confidential company data	Keeping trade secrets
Whistleblower Situations	Reporting illegal activities	Reporting fraud
Compromising Security	Weakening security to reduce costs	Removing firewall rules
Consultation Fees	Charging excessive fees by exploiting fear	Overselling security services
Slippery Slope	Small unethical actions lead to larger ones	Reading emails just once”

Organizational Ethical Policies

Policy	Purpose	Example
Informed Consent	Users know the rules	Company monitoring policy
User Code of Conduct	Define acceptable use	Internet usage policy
Privileged Access Code of Conduct	Control administrator privileges	Root account rules
Backups	Recover from mistakes	Daily backups
Retain Software Sources	Recover source code	Git repository

LOPSA Code of Ethics

Principle	Purpose
Professionalism	Maintain professional standards
Personal Integrity	Be honest
Privacy	Respect confidentiality
Laws & Policies	Follow legal requirements
Communication	Communicate honestly
System Integrity	Protect systems
Education	Continue learning
Responsibility to Computing Community	Support the profession
Social Responsibility	Benefit society
Ethical Responsibility	Always act ethically

Chapter 6

Causes of Network Security Breaches

Cause	How it Happens	Example	Prevention
Human error	Employees accidentally create vulnerabilities	Installing pirated software infected with malware	Security awareness training
Unsecured workstations	Users leave computers unlocked	Attacker accesses an unlocked PC	Lock computers when away
Insider information leakage	Employees reveal sensitive information	Sharing credentials with outsiders	Security policies & training
Physical access	Attacker gains access to network devices	Extracting data from servers	Physical security controls
Security holes	Vulnerabilities in OS, network devices, or protocols	Exploiting an unpatched OS	Updates and patch management

Types of Cybercrime

Cybercrime	Definition	Primary Goal	Example
Computer Fraud	Deception for personal gain	Financial benefit	Fake online identity
Cyberterrorism	Politically motivated attack causing major harm	Disruption	Attack on power grid
Hactivism	Political/social hacking without severe damage	Protest	Website defacement
Cybertrespass	Unauthorized access	Access	Breaking into a server
Cyber-vandalism	Damaging systems or data	Destruction	Corrupting databases
Cyber-piracy	Illegal copying/distribution	Copyright infringement	Cracked software

Types of Network Security Attacks

Attack	How it Works	Main Target	Example	Prevention	Keyword
Unstructured Attack	Unskilled hackers using existing tools	General systems	Script kiddie	Basic security	Low skill
Structured Attack	Skilled hackers create custom attacks	Vulnerable systems	Professional penetration	Patch vulnerabilities	High skill
Social Engineering	Manipulates people	Humans	Fake IT support	Awareness	Human trust
Password Attack	Cracks passwords	User accounts	Brute-force login	Strong passwords	Password
Dictionary Attack	Uses common words/patterns	User accounts	Guessing “password123”	Complex passwords	Dictionary
Exploit Attack	Uses software bugs	Software	Buffer overflow	Software updates	Vulnerability
Logic Attack	Hidden malicious code	Software	Logic bomb	Code review	Trigger
Eavesdropping	Listens to traffic	Network communication	Packet sniffing	Encryption	Sniffing
Compromised-Key Attack	Stolen private key decrypts traffic	Encrypted communication	Key theft	Protect keys	Private key
Man-in-the-Middle (MITM)	Intercepts and modifies communication	Network traffic	Altering messages	Encryption & authentication	Intercept
IP Address Spoofing	Pretends to be another device	Network identity	Fake IP	Filtering & authentication	Fake IP
Application-Layer Attack	Exploits applications	Servers/Workstations	Trojan infection	Secure applications	Applications
Resource Attack	Exhausts network resources	Network availability	Flooding requests	Traffic filtering	Resources
DoS	One attacker overwhelms server	Availability	Server flooding	Rate limiting	One source
DDoS	Many compromised devices attack together	Availability	Botnet	DDoS protection	Many sources
Email Bombing	Floods mailbox with spam	Email service	Thousands of emails	Spam filtering	Inbox flood
Backdoor	Hidden access mechanism	Authentication	Secret admin account	Auditing	Hidden access
Time Bomb	Activates after a condition/time	Entire system	Admin leaves company	Code audits	Delayed trigger
Stealth Backdoor	Hidden backdoor that avoids detection	Administrator access	Remote hidden account	Logging & audits	Invisible access

Network Security Countermeasures

Countermeasure	Purpose	Protects Against
Security Policy	Establish security rules	Human threats
Firmware & OS Updates	Fix vulnerabilities	Exploit attacks
Strong Passwords	Prevent password attacks	Password/Dictionary attacks
Account Lockout	Stop repeated guessing	Brute force
One-Time Passwords	Prevent spoofing	IP Spoofing
Two-Step Verification	Strengthen authentication	Account compromise
Antivirus	Detect malware	Application-layer attacks
Firewall	Filter network traffic	External attacks
Logging Software	Detect suspicious activity	Backdoors
Surprise Audits	Detect insider threats	Stealth backdoors
Smart Cards	Secure authentication	Credential theft
Biometrics	Identity verification	Unauthorized access
Self-Policing	Promote ethical conduct	Insider threats
Ethical Behaviour	Build security culture	Human threats
Zero Tolerance	Deter violations	Insider attacks
Adequate Resources	Prevent admin overload	Security mistakes

Chapter 7

All the Technologies Used to Collect and Track Personal Information

Technology	Purpose	How It Works	Privacy Concern	Example	Keyword
Cookies	Remember user preferences and track browsing	Website stores a small file on your device	Tracks browsing history; advertising companies may use the data	Amazon remembers your cart	Stored file
Flash Cookies (Supercookies)	Persistent tracking	Stored outside normal browser cookies	Difficult to delete using normal browser settings	Website continues recognizing you after clearing cookies	Persistent
Device Fingerprinting	Identify a device without cookies	Collects browser, fonts, OS, clock, hardware settings	Invisible tracking that is difficult to detect or block	Website identifies your laptop without storing cookies	Unique device
Cross-device Tracking	Link activity across multiple devices	Connects data from phone, tablet, and computer	Builds one profile across all devices	Ads follow you from your phone to your laptop	Multiple devices
Online Behavioral Tracking	Monitor browsing behavior	Advertising networks observe activity across websites	Creates detailed behavioral profiles without users realizing	Seeing ads based on websites you visited	Behavior profile
RFID	Identify and track objects or people	Uses a microchip (tag) and radio waves read by a scanner	Objects or people can be tracked without their knowledge	RFID passport or employee badge	Radio tag
Dataveillance	Monitor people's activities	Cameras, workplace software, surveillance systems	Constant monitoring threatens privacy	CCTV or employee monitoring software	Continuous surveillance

Information Processing Techniques

Technique	What It Does	Purpose	Example	Privacy Concern
Data Mining	Extracts patterns from large datasets	Predict behavior	Shopping habits predict advertisements	Profiling and discrimination
Computer Merging	Combines information from multiple databases into one profile	Build a complete profile	Bank + hospital + shopping records	Violates contextual integrity if done without permission
Computer Matching	Cross-checks different databases to find matches (hits)	Detect fraud or suspects	Government compares tax and welfare records	User did not consent to sharing between agencies

Privacy Technologies

Technology	Main Function	Advantage	Privacy Risk
Cookies	Personalization	Better user experience	Tracks browsing
Flash Cookies	Persistent storage	Remember preferences longer	Hard to remove
Fingerprinting	Device identification	Doesn't rely on cookies	Nearly impossible to detect
RFID	Object identification	Fast identification	Physical tracking

Privacy Concepts

Concept	Definition	Focus	Example
Privacy	Right to be left alone	Freedom from intrusion	Nobody reads your diary
Information Privacy	Control over personal information	Collection, use, sharing	Choosing who sees your medical records
Cyber Privacy	Privacy in digital environments	Online data	Protecting browsing history

Data Collection Methods

Method	What Is Collected	Example
Social Media	Posts, likes, profile information	Facebook
Mobile Devices	GPS, sensors, cameras	Smartphone location
IoT	Smart device statistics	Smart thermostat
Biometrics	Fingerprints, face, iris, voice	Face ID
E-Government	Citizen records	Digital passport

Privacy Protection Methods

Protection	Purpose	Example
Antivirus	Prevent malware	Windows Defender
Firewall	Block unauthorized access	Windows Firewall
Encryption	Protect stored/transmitted data	HTTPS
P3P	Compare website privacy policies with user preferences	Browser privacy settings
Privacy Awareness	Educate users	Training and best practices

Security Issues in Cloud Computing

Security Issue	Definition	Main Target	Main Cause	Main Consequence	Example	Keyword
Data Breach	Unauthorized access to confidential information	Data	Hackers, human error, vulnerabilities, poor security	Information is exposed	Hacker steals patient records	Exposed data
Account Hijacking	Attacker steals user credentials	User account	Stolen passwords, phishing	Attacker acts as the legitimate user	Hacker logs into your Gmail	Credentials
Malware Injection	Malicious code is inserted into cloud services as a valid instance	Cloud service	Injected malicious code	Malware executes inside the cloud	Fake SaaS application	Valid instance
Shared Vulnerabilities	Security is shared between provider and client	Both provider & client	Weak passwords, poor client security	Increased risk of attack	Client doesn't enable MFA	Shared responsibility
Data Loss	Data is permanently destroyed or becomes unrecoverable	Data	Attack, disaster, provider error	Data disappears	Provider accidentally deletes database	Permanent loss

Ethical Issues in Cloud Computing

Ethical Issue	Definition	Main Concern	Example	Related Ethical Theory	Keyword
Compliance	Following standards and regulations	Following rules	Provider follows ISO standards	Professional Ethics	Standards
Spiteful Activity	Insider intentionally causes harm	Revenge	Fired employee deletes files	Professional Ethics	Insider
Policies Ignoring Customer Interests	Policies favor provider over customers	Fairness	Selling user data	Utilitarianism	Customers
Intellectual Property	Illegal sharing of digital content	Ownership	Uploading movies	Copyright	Copyright
Discrimination	Unfair treatment of customers	Equality	Prioritizing big companies	Justice/Fairness	Bias

Legal Issues

Legal Issue	Definition	Main Concern	Example	Keyword
Jurisdiction	Which country's laws apply	Country	Data stored in Germany	Laws
Data Responsibility	Responsibility for stored data	Liability	Provider must protect data	Responsibility
Third-party Access	Partners accessing data	Privacy	Outsourcing	Third Party
Rights Over Files	Ownership after leaving provider	Ownership	Company switches providers	Ownership
Terms & Conditions	Users rarely read agreements	Consent	Accepting terms	Agreement
National Records	Special laws for government data	Government	National ID database	Government

Social Issues

Social Issue	Problem	Who is Affected	Example	Keyword
High Cost	Expensive resources	Everyone	Buying more storage	Cost
Affordability	Some cannot pay	Individuals & SMEs	Startup can't afford cloud	Money
Unequal Competition	Small companies can't compete	Small businesses	Amazon vs Startup	Competition
Unequal Support	Large customers get priority	Small customers	Premium support	Fairness

Chapter 8

Applications of Social Media

Application	Purpose	Example	Benefit
Employee Hiring	Recruit and screen candidates	LinkedIn recruiting	Better hiring decisions
Organizations	Internal communication & collaboration	Company social platform	Better teamwork
Digital Marketing	Promote products and brands	Facebook campaigns	Increased sales

Social Media in the Hiring Process

Use	Description	Benefit	Ethical Concern
Recruitment	Advertise vacancies and attract candidates	Larger talent pool	Fair hiring practices
Background Checks	Review candidates' online profiles	Verify qualifications	Privacy, discrimination

Information That Helps vs Hurts a Candidate

Positive Information	Negative Information
Volunteer work	Drug or alcohol use
Professional achievements	Inappropriate photos
Certifications	Discriminatory remarks
Positive online reputation	Sharing confidential information

Social Media in Organizations: Pros vs Cons

Pros	Cons
Better communication	Fraud and hacking
Knowledge sharing	Scams and identity theft
Wider business contacts	Malware and viruses
Better recruitment	Negative employee comments
Improved reputation	Legal liability
Better marketing	Reduced productivity

Social Networking Use Policy at Work

Policy Element	Purpose
Define social networking	Employees know what is covered
State policy purpose	Clear organizational expectations
Explain benefits	Encourage responsible use
Employee education	Improve awareness
Security awareness	Prevent cyberattacks
Safe browsing guidance	Avoid suspicious links
Protect personal information	Reduce security risks

Role of Social Media in Digital Marketing

Benefit	Description
Target customers	Understand customer needs
Immediate response	Quickly solve customer problems
Increase sales	Reach more customers
Low cost	Free or inexpensive marketing
Brand recognition	More people know the company
Brand loyalty	Build customer relationships
Customer insights	Learn customer preferences

Brand Promotion on Social Media: Benefits vs Drawbacks

Benefits	Drawbacks
Better social signals (SEO)	Competitors observe strategies
Brand awareness	Need qualified personnel
Word-of-mouth marketing	Brand reputation can be damaged
Customer insights	Time consuming
Better customer service	Requires continuous management
Cost efficient	Negative comments spread quickly
Connectivity	Ongoing monitoring required
More sales	Resource intensive

Safe Social Networking Practices

Recommendation	Purpose
Use strong unique passwords	Prevent account compromise
Share minimal personal information	Protect privacy
Configure privacy settings	Control visibility
Disable unnecessary third-party apps	Prevent data leakage
Think before posting	Protect reputation
Don't post employer information	Protect company confidentiality
Supervise children's accounts	Improve child safety
Be suspicious of links and requests	Prevent phishing
Avoid random ads and downloads	Reduce malware risk
Use browser security tools (WOT, NoScript, Adblock, anti-phishing)	Detect malicious websites
Google yourself	Monitor online reputation
Think before clicking	Avoid scams and malware

Chapter 9

Introduction to Ethical Hacking

Information Security Objectives (CIA + Non-Repudiation)

Principle	Goal	If Lost	Common Protection
Confidentiality	Prevent unauthorized disclosure	Privacy breach	Encryption, Authentication, Access Control
Integrity	Prevent unauthorized modification	Incorrect/Fake data	QA, Audit Logs, Checks
Availability	Ensure systems are accessible	Downtime	Backup, Redundancy, Capacity Planning
Non-repudiation	Prevent denying an action	Sender denies transaction	Digital Signatures, Encryption

Security Terminologies

Term	Purpose	Key Idea
Vulnerability	Weakness	Entry point
Exploit	Uses vulnerability	Gains access
Zero-Day Attack	Exploits unknown flaw	No patch exists
Hack Value	Target attractiveness	High-value target
Payload	Malicious action	Executes attack
Bot	Automated malware	Controlled remotely
Botnet	Group of bots	Large-scale attacks
Daisy Chaining	Multiple linked attacks	One attack enables another
Doxing	Publish private info	Uses public sources

Cyber Attack Components

Component	Definition	Example
Motive	Why attack	Money, espionage
Method	How attack	Phishing, Malware
Vulnerability	Weakness exploited	Unpatched software

Modern Cyber Threats

Threat	Main Goal	Example
Cloud Threats	Attack cloud resources	Misconfigured cloud
APT	Long-term espionage	Government attack
Virus/Worm	Spread malware	Infected attachment
Mobile Threats	Attack smartphones	Spyware
Insider Threat	Authorized misuse	Employee steals data
Botnet	Coordinated attacks	DDoS

Mobile Threats

Threat	Description
Data Leakage	Sensitive data exposed
Phishing	Fake messages
Spyware	Secret monitoring
Broken Cryptography	Weak encryption
Network Spoofing	Fake network
Unsecured Wi-Fi	Public network attacks

Threat Categories

Network Level	Host Level	Application Level
Scanning	Malware	SQL Injection
Sniffing	Password Attacks	Input Validation
Spoofing	Login Bypass	Authentication Failures
Session Hijacking	Privilege Escalation	Session Management
MITM	Backdoors	Buffer Overflow
DNS/ARP Poisoning	Arbitrary Code Execution	Cryptography Failures
		Misconfiguration
		Error Handling

Information Warfare

Defensive	Offensive
Protect systems	Attack systems
Monitoring	Disruption
Security controls	Manipulation
Incident response	Destroy infrastructure

Social Engineering

Common Social Engineering Attacks

Attack	Method	Target
Phishing	Email	Anyone
Spear Phishing	Targeted email	One person
Whaling	Executive phishing	CEO/Managers
Vishing	Phone	Victims
Smishing	SMS	Mobile users
Impersonation	Pretend to be someone	Employees
Dumpster Diving	Trash search	Organizations

Phishing vs Spear Phishing vs Whaling

Feature	Phishing	Spear Phishing	Whaling
Target	Everyone	One person	Executives
Personalization	Low	High	Very High
Goal	Credentials	Specific victim	High-value access

Phishing vs Vishing vs Smishing

Attack	Medium	Typical Example
Phishing	Email	Fake login page
Vishing	Phone	Fake bank call
Smishing	SMS	Fake prize/bank text

Impersonation Attacks

Type	Goal	Example
Delivery Person	Physical access	Fake courier
Tech Support	Computer access	Fake IT technician

Protecting Against Social Engineering

Protection	Prevents
Strong spam filters	Phishing
Verify URLs	Fake websites
Never share passwords	Credential theft
Ignore urgent requests	Scams
Update devices	Malware
Anti-virus	Malware
Anti-phishing tools	Fake websites
Verify unknown callers	Vishing
Security awareness training	All attacks

Business Ethics

Types of Ethics

Type	Focus	Example
Personal Ethics	Individual values	Personal honesty
Professional Ethics	Profession	Accountant ethics
Organizational Ethics	Company values	Company culture
Global Ethics	International business	Respect local customs

Ethics vs Law

Ethics	Law
Moral principles	Legal requirements
Voluntary	Mandatory
Beyond legal minimum	Minimum acceptable behavior
May be legal but unethical	Enforced by government

Core Ethical Principles

Principle	Meaning
Integrity	Honesty
Fairness	Equal treatment
Respect	Protect dignity
Responsibility	Accountability
Transparency	Open communication

CSR vs Code of Ethics vs Whistleblowing

Concept	Purpose
Corporate Social Responsibility (CSR)	Benefit society
Code of Ethics	Guide employee behavior
Whistleblowing	Report unethical acts

Ethical Decision-Making Model

Step	Action
1	Recognize issue
2	Gather facts
3	Apply ethical principles
4	Evaluate alternatives
5	Reflect on outcome

Benefits vs Consequences of Ethics

Ethical Organization	Unethical Organization
Customer trust	Reputation loss
Brand loyalty	Lawsuits
Transparency	Employee dissatisfaction
Long-term success	Financial losses
Sustainability	Market exit

Chapter 10

The Four Types of Intellectual Property

Type	Protects	Examples	Duration
Copyright	Creative works	Books, music, software	Author's life + 50 years (Saudi slides)
Patent	Inventions	Machine, process	20 years
Trademark	Brand identity	Logo, name, slogan	Unlimited (renewable)
Trade Secret	Confidential business information	Coca-Cola formula	As long as it remains secret

Copyright vs Patent vs Trademark vs Trade Secret

Feature	Copyright	Patent	Trademark	Trade Secret
Protects	Creative expression	Inventions	Brand identity	Confidential information
Registration Required	No	Yes	Yes (recommended)	No
Prevents Copying	✓	✓	✓	✓
Prevents Independent Creation	✗	✓	✗	✗
Time Limit	Life + 50 years	20 years	Renewable forever	Unlimited while secret

What Copyright Covers

Protected	Not Protected
Books	Ideas
Music	Common knowledge
Software	Intangible ideas
Movies	Facts
Architecture	Concepts

Four Copyright Rights

Right	Meaning	Example
Reproduction	Copy work	Photocopying book
Derivative	Create modified work	Translate book, movie from novel
Performance	Public display/performance	Concert, theater
Distribution	Sell/rent/give copies	Selling DVDs

Copyright Infringement Requirements

Requirement	Explanation
Valid copyright	Owner owns copyright
Access	Infringer had access
Unauthorized copying	Copy exceeds legal exceptions

Fair Use

Allowed	Not Automatically Allowed
Teaching	Commercial copying
Research	Large-scale copying
News reporting	Unpublished works
Criticism	Copying entire work

Fair Use Factors

Factor	Question
Purpose	Commercial or educational?
Nature	What type of work?
Amount	How much copied?
Market Effect	Does it reduce sales?

Software Copyright vs Software Patent

Copyright	Patent
Protects source code	Protects invention/process
Automatic	Requires application
Doesn't protect ideas	Protects functional invention
Short snippets may not qualify	One feature may be patented

DRM vs DMCA

DRM	DMCA
Technology	Law
Encrypts content	Protects DRM
Restricts access	Makes bypassing DRM illegal
Example: Kindle books	Prevents hacking DRM

Plagiarism vs Copyright Infringement

Plagiarism	Copyright Infringement
Claims another's work as yours	Uses copyrighted work without permission
Ethical issue	Legal issue
Can happen with permission	Can happen even with attribution
Mainly academic	Applies everywhere

Patent Requirements

Requirement	Meaning
Legal class	Process, machine, manufacture, composition, improvement
Useful	Has practical use
Novel	New invention
Non-obvious	Not obvious to skilled person

Cannot patent:

- Abstract ideas
- Laws of nature
- Natural phenomena

Copyright vs Patent

Copyright	Patent
Creative work	Functional invention
Automatic	Application required
Independent creation allowed	Independent creation prohibited
Life + 50 years	20 years

Patent Infringement Types

Type	Description
Direct	Personally make/use/sell invention
Indirect	Help infringement
Induced	Encourage others
Contributory	Supply special components
Literal	Exact patented invention copied
Willful	Intentional infringement

Trademark

Trademark Protects	Examples
Logos	Apple logo
Words	Nike
Symbols	™
Sounds	Intel sound
Colors	Tiffany Blue
Shapes	Coca-Cola bottle
Trade Dress	Product packaging/look

Trademark vs Copyright

Trademark	Copyright
Brand identity	Creative work
Renewable forever	Limited duration
Prevents consumer confusion	Prevents copying
Registration preferred	Automatic

Trade Secret Requirements

Requirement	Explanation
Economic value	Valuable
Cost/Effort	Difficult to create
Unique	Some novelty
Secret	Not public
Protected	Company actively protects it

Trade Secret vs Patent

Trade Secret	Patent
Secret	Public
Unlimited duration	20 years
No registration	Registration required
Lost if disclosed	Protected by law

NDA vs Non-Compete

NDA	Non-Compete
Don't reveal secrets	Don't work for competitors
Protect confidentiality	Protect market advantage

Cybersquatting

Definition	Goal	Prevention
Register similar domain	Profit from trademark	Register multiple domains

Common examples:

- Different extension (.com → .net)
- Misspelled domain
- Sell domain later
- Advertising using famous brand

Competitive Intelligence vs Reverse Engineering

Competitive Intelligence	Reverse Engineering
Gather public information	Take product apart
Business strategy	Understand/copy/improve
Ethical information gathering	Analyze hardware/software

Saudi IP Facts

Topic	Fact
Saudi Copyright	Royal Decree M/41 (2003)
Copyright Duration	Life + 50 years
SAIP	Saudi Authority for Intellectual Property
Purpose of SAIP	Manage and enforce IP in Saudi Arabia

Chapter 11

Classification of Cyber Crimes

Category	Target	Examples
Cyber against Individual	Individuals	Email Spoofing, Spamming, Phishing, Cyber Stalking, Cyber Defamation, Cyber Pornography
Cyber against Property	Property and digital assets	Credit Card Skimming, Intellectual Property Crimes, Software Piracy, Identity Theft, DDoS, Hacking, Virus Transmission, Cybersquatting, Copyright Infringement, IPR Violations
Cyber against Government	Government systems	DoS, Email Bombing, Logic Bomb, Data Diddling, Sale of Illegal Articles, Cyber Terrorism, Pirated Software, Accessing Confidential Information

Saudi Anti-Cybercrime Law Overview

Item	Details
Royal Decree	M/17
Date Issued	26 March 2007
Number of Articles	16
Purpose	Combat cybercrime in Saudi Arabia

Important Cyber Crimes

Crime	Description	Example
Data Diddling	Unauthorized alteration of data before or during processing, then changing it back after processing	Changing a customer's delivery address before processing
Salami Attack	Stealing tiny amounts from many victims so each loss goes unnoticed	Deducting SAR 2 from every bank account
Trojan (Keylogger)	Records every keystroke to steal sensitive information	Capturing online banking passwords
Web Jacking	Taking control of a website by changing login credentials	Hijacking a company website for ransom
Credit Card Skimming	Stealing credit card information during transactions	Hidden skimming device on a card reader
Identity Theft	Using another person's identity for fraud	Opening a bank account using stolen identity
Email Bombing	Flooding an inbox with emails until it becomes unusable	Sending thousands of emails to one address
Logic Bomb	Malicious code that activates when a specific condition is met	Program deletes files when an employee leaves
Email Spoofing	Forging the sender's email address	Fake email appearing to come from a bank
Cyber Stalking	Using the Internet to harass or threaten someone	Repeated threatening emails
Cyber Defamation	Publishing false statements online to damage someone's reputation	Posting false accusations on social media

Objectives of the Saudi Anti-Cybercrime Law

Objective	Purpose
Enhancement of Information Security	Protect computer systems and networks
Protection of Legitimate Use of Computers and Networks	Safeguard users' rights
Protection of Public Interest, Morals, and Common Values	Preserve society and public order
Protection of the National Economy	Reduce economic damage caused by cybercrime

Saudi Anti-Cybercrime Law (Articles 3–7)

Article	Main Crimes	Maximum Prison	Maximum Fine
Article 3	Spying, interception of data, blackmail, website hacking, invasion of privacy, cyber defamation	1 year	500,000 SAR
Article 4	Fraud, false identity, illegal access to banking and credit information	3 years	2 million SAR
Article 5	Deleting, destroying, leaking, altering data, disrupting networks and services	4 years	3 million SAR
Article 6	Content against public morals, pornography, gambling, human trafficking, narcotics	5 years	3 million SAR
Article 7	Terrorism, threatening national security or the national economy	10 years	5 million SAR

Article 9

Situation	Punishment
Incites, assists, or collaborates in committing a cybercrime	Up to the same maximum punishment as the offender if the crime is committed
Intended crime is not completed	Up to half of the maximum punishment

Commonly Confused Cyber Crimes

Crime	Key Difference
Phishing	Tricks victims into revealing sensitive information
Email Spoofing	Fakes the sender's identity
Spamming	Sends unsolicited bulk emails
Email Bombing	Overwhelms a mailbox or mail server with emails
DoS	Prevents legitimate users from accessing a service
Data Diddling	Alters data before or during processing
Salami Attack	Steals very small amounts from many victims
Identity Theft	Uses another person's identity
Credit Card Skimming	Steals payment card information
Keylogger	Records keyboard input
Web Jacking	Takes control of a website